

Fair Cohort Selection in University Admissions

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Abstract

We study the problem of fair cohort selection in university admissions, formulating the decision process as an Integer Linear Program (ILP) subject to demographic constraints. We extend the standard utility-maximization baseline to enforce five distinct fairness notions: Demographic Parity, Equality of Opportunity, Intersectional Demographic Parity, Max-Min Fairness, and α -Fairness. We then apply these models to a synthetic dataset based on the Wharton MBA Class of 2025 to evaluate the trade-off between institutional utility and equity. Our empirical results confirm this expected trade-off. We also observe that enforcing fairness on single attributes (e.g., race) fails to resolve intersectional disparities. Surprisingly, we find that the unconstrained merit-based baseline exhibits lower intersectional disparity than actual human decision-making. Theoretically, we propose a reformulation of relative disparity constraints into integer cardinality bounds to preserve the Total Unimodularity of the constraint matrix, ensuring that the Linear Programming relaxation admits integral solutions.

1 Introduction & Background

Consider the following selection problem: Given a pool of n applicants, each with a vector of attributes (e.g., GPA, test scores) and membership in one or more protected groups (e.g., race, gender), the task is to select a cohort of exactly k students. The baseline optimization problem seeks to maximize an “institutional utility” function, such as the total academic merit of the incoming class, subject to the class size constraint.

The fundamental problem is that a purely utility-maximizing selection can result in significant demographic disparities, violating institutional or societal fairness goals. In this project, we formulate an Integer Linear Program with explicit fairness constraints. Consider the following baseline model:

$$\begin{aligned} & \text{maximize} && \sum_{i=1}^n u_i x_i \\ & \text{subject to} && \sum_{i=1}^n x_i = k \\ & && x_i \in \{0, 1\} \quad \forall i = 1, \dots, n \end{aligned} \tag{1}$$

Here, x_i is a binary decision variable (=1 if applicant i is admitted and 0 otherwise) and u_i represents the utility or merit score of applicant i indicating their academic performance (e.g., GPA/GMAT).

Contribution.¹ We extend this baseline ILP to enforce five common notions of *group fairness* — Demographic Fairness, Equality of Opportunity, Intersectional Demographic Parity, Max-Min Fairness, and α -Fairness. We will then apply this model to a real-world university admissions dataset, the MBA dataset (Wharton, 2024) and evaluate the efficiency-fairness trade-off.

2 Previous Work

This project is situated at the confluence of three established research streams. The first is the application of optimization to higher education operations and specifically admissions. This includes work on using linear programming to allocate merit-based financial aid to maximize the academic quality (e.g., average SAT scores) of an incoming class. More recent work has proposed integer programming (IP) models to formalize the student selection process as a complex “constrained assignment problem” (Majdoub et al., 2024). These models aim to balance institutional goals (like academic merit) with student preferences and resource constraints.

¹We provide all code and data necessary to reproduce our work at https://github.com/esampi/fair_cohort_selection.

The second stream consists of methodological papers on formulating fairness criteria for algorithms as mathematical constraints (Chen and Hooker, 2023). We study IP formulations of the well-known fairness principles of Demographic Parity and Equalized Odds (Hardt et al., 2016; Dwork et al., 2012) to the practical application context of university admissions. Moreover, we also look at fairness principles that stem from network flow problems, such as Max-Min fairness (Bertsekas and Gallager, 1992) and proportional or α -fairness (Kelly et al., 1998; Mo and Walrand, 2002).

The third stream combines the first two streams to study "fair cohort selection" (Dwork and Ilvento, 2019; Bairaktari et al., 2022; Kleinberg et al., 2018). This area specifically studies the problem of selecting a subset of size k from a larger pool, with the objective of maximizing some utility function subject to fairness constraints. This work has produced various optimization frameworks, such as sequential linear programs (LPs) to achieve leximax fairness (Bairaktari et al., 2022; Henzinger et al., 2022) or integer linear programs to enforce diversity constraints.

3 Fairness Notions

We start by giving an overview of the most common fairness notions in algorithmic fairness.

Fairness Notion 1: Demographic Parity. This model enforces *Demographic Parity* (DP, also known as statistical parity), which requires that the selection rate (probability of admission) be equal across all protected groups (Dwork et al., 2012). Let N_g be the set of applicants belonging to a protected group $g \in G$. The selection rate for group g is $w_g = \frac{1}{|N_g|} \sum_{i \in N_g} x_i$. The model minimizes the maximum disparity ϵ between any two groups, or, more simply, solves 1 for a range of ϵ values with additional constraints. We summarize all constraints in Table 1.

Fairness Notion 2: Equality of Opportunity. This model enforces *Equality of Opportunity* (EO), a component of *Equalized Odds* (Hardt et al., 2016). This notion requires that the selection rate be equal *only among qualified applicants*. This formulation requires a binary (or categorical) "ground truth" proxy for qualification, a_i , where $a_i = 1$ if applicant i is considered qualified and 0 otherwise. Let N_g^q be the set of qualified applicants in group g (i.e., $i \in N_g$ and $a_i = 1$). The true positive rate (TPR) for group g is $w_g^q = \frac{1}{|N_g^q|} \sum_{i \in N_g^q} x_i$.

Fairness Notion 3: Intersectional Demographic Parity. To capture disparities that arise across combinations of protected attributes, we extend Demographic Parity to *intersectional groups* (e.g., Black women, White men). Let the protected attributes be race $r \in R$ and gender $g \in G$, and define each intersectional group as $h = (r, g) \in H := R \times G$. Let N_h be the set of applicants in group h . The intersectional selection rate is $w_h = \frac{1}{|N_h|} \sum_{i \in N_h} x_i$. The model is defined as in 1 with the additional model constraints:

This extension allows us to evaluate whether fairness enforced on individual attributes (e.g., race or gender) propagates to their intersections, a property that is often violated in practice.

Fairness Notion	Rate Definition	Constraint
Demographic Parity (DP)	$w_g = \frac{1}{ N_g } \sum_{i \in N_g} x_i$	$ w_g - w_{g'} \leq \epsilon \quad \forall g, g' \in G$
Equality of Opportunity (EO)	$w_g^q = \frac{1}{ N_g^q } \sum_{i \in N_g^q} x_i$	$ w_g^q - w_{g'}^q \leq \epsilon \quad \forall g, g' \in G$
Intersectional DP (IDP)	$w_h = \frac{1}{ N_h } \sum_{i \in N_h} x_i$	$ w_h - w_{h'} \leq \epsilon \quad \forall h, h' \in H$

Table 1: Summary of fairness metrics and constraints.

Aside from the conventional algorithmic fairness definitions, there exist a range of other fairness principles common in network flow problems that can be applied to our setting. We provide a short overview in the following:

Fairness Notion 4: (Regularized) Max-Min (Rawlsian) Fairness. We consider fairness from a Rawlsian perspective, where the goal is to maximize the welfare of the least well-off group (Bertsekas

and Gallager, 1992). In our context of university admissions, this implies maximizing the minimum selection rate across all groups:

$$\text{maximize } z + \delta \sum_{i=1}^n u_i x_i \quad \text{s.t.} \quad \frac{1}{|N_g|} \sum_{i \in N_g} x_i \geq z \quad \forall g \in G \quad (2)$$

In contrast to the previous approaches, max-min fairness does not use a fairness budget ϵ ; rather, it represents the extreme point of the efficiency-fairness trade-off, forcing acceptance rates to be equal across groups. One benefit this formulation has over the Demographic Parity formulation (3) is that while DP might be infeasible for small ϵ , 2 always returns a feasible solution (the integer allocation that is mathematically closest to perfect equality within integer rounding limits). The regularization term ($\delta \sum u_i x_i$) acts as a tie-breaker, ensuring that within the strict fairness requirements, the highest-utility candidates are still prioritized.

Fairness Notion 5: α -Fairness (Proportional Fairness). Instead of relying on a fairness budget ϵ , we can also employ the concept of α -fairness to bridge the gap between pure utility maximization (Baseline) and pure equity (Max-Min). This creates a spectrum of objectives parameterized by $\alpha \geq 0$, maximizing the sum of isoelastic functions of group utility $U_g = \sum_{i \in N_g} u_i x_i$:

$$\text{maximize} \quad \sum_{g \in G} \frac{(U_g)^{1-\alpha}}{1-\alpha}$$

- $\alpha = 0$ (**Utilitarian**): Equivalent to the baseline model; 1 unit of utility for a well-represented group is equal to 1 unit for an under-represented group.
- $\alpha \rightarrow 1$ (**Proportional**): Converges to $\sum \log(U_g)$ (Nash Bargaining). This penalizes inequality by introducing diminishing returns—adding utility to a group that already has high representation yields little objective gain.
- $\alpha \rightarrow \infty$ (**Max-Min**): Converges to equal group utilities, similar to the Rawlsian objective discussed above.

While this formulation guarantees feasibility (unlike hard constraints which may be infeasible for small ϵ), the objective function is concave. To solve this with MILP solvers, we approximate the objective using piecewise linear functions.

4 A Note about Ideal Formulations

Our baseline is the efficiency-maximal (unconstrained) solution of 1. Its optimal solution is trivial to find: it corresponds to admitting the k applicants with the highest utility scores u_i . It has the maximum possible utility (0% "Price of Fairness") and the maximum (worst) fairness violation.

We can rewrite the equality constraint in the baseline model as $Ax = k$, where $A = [1, \dots, 1]$ is the vector of n ones. Naturally, since A has the "consecutive ones" property, it is Totally Unimodular (TU). Therefore, per Theorem 4.3 in the lecture notes (Iancu, 2025), the LP relaxation of 1 admits an integral optimal solution.²

However, adding ratio-based constraints such as $|w_g - w_{g'}| \leq \epsilon$ (where $w_g = \frac{1}{|N_g|} \sum_{i \in N_g} x_i$, $\forall g \in G$) destroys this property. The resulting polyhedron for the LP relaxation may contain fractional vertices, requiring computationally more expensive IP solvers (like Branch-and-Bound) to find integer solutions. Even though it is computationally feasible to use IP solvers for our problem, and we use those for the remainder of this report, we still want to outline an alternative, equivalent formulation (**an ideal formulation**) of our optimization problem that admits an integral LP relaxation.

4.1 Formulation with Cardinality-Based Constraints

To recover the TU property for disjoint protected groups, we replace the relative disparity constraints ($|w_g - w_{g'}| \leq \epsilon$) with absolute cardinality bounds. This structure aligns with network flow problems, which guarantee integral solutions.

²We assume that the utility distribution is not degenerated.

Let $r_{global} = k/n$ be the global acceptance rate. To ensure that the acceptance rate of any group g does not deviate from the global average by more than $\epsilon/2$ (ensuring a maximum pairwise disparity of ϵ), we convert these rates into integer lower bounds (L_g) and upper bounds (U_g) as follows (see Appendix A for a derivation):

$$L_g = \left\lceil |N_g| \left(\frac{k}{n} - \frac{\epsilon}{2} \right) \right\rceil, \quad U_g = \left\lfloor |N_g| \left(\frac{k}{n} + \frac{\epsilon}{2} \right) \right\rfloor$$

The optimization problem effectively becomes a resource allocation problem with cardinality constraints:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^n u_i x_i & \text{subject to} \quad & \sum_{i=1}^n x_i = k \\ & & & L_g \leq \sum_{i \in N_g} x_i \leq U_g \quad \forall g \in G \\ & & & 0 \leq x_i \leq 1 \end{aligned} \tag{3}$$

Because the columns associated with each group N_g are disjoint, the constraint matrix for 3 is an interval matrix (up to permutation of columns), guaranteeing that the optimal solution to the LP relaxation is binary.

4.2 Granularity and Limitations

While this formulation is computationally efficient, it introduces a "granularity" challenge. Whereas in our fractional formulation, ϵ can be continuous, in the integer domain, the minimum feasible step size for a group's selection rate is $1/|N_g|$. For small groups (e.g., indigenous groups or intersectional groups), this step size may exceed the fairness budget ϵ . If $\epsilon < 1/|N_g|$, the bounds may tighten such that $L_g = U_g$ (forcing exact proportionality), a risk noted in our proposal regarding trivial solutions.

5 Data & Implementation

5.1 Data

We now apply our fair optimization problem to the **MBA dataset**³ (Wharton, 2024). The MBA dataset contains 6,194 applicant profiles synthetically generated from true statistics of Wharton's 2025 MBA class, and each applicant profile includes their academic profile (GPA, GMAT, work experience, work industry, undergraduate major) and protected attributes (race, gender).

5.2 Defining Institutional Utility

We define **institutional utility** as the value an institution seeks to maximize in a hypothetical unconstrained environment—specifically, in the absence of fairness mandates, diversity goals, or regulatory requirements. While a university's true unconstrained objective might prioritize long-term outcomes unobservable to us (e.g., "select the class that will generate the most Alumni donations in 30 years"), we restrict our focus to observable metrics that serve as proxies for academic excellence and institutional reputation.

It is important to note that this utility metric represents a theoretical "pure merit" baseline, and is distinct from actual historical admission decisions, which likely *did* incorporate fairness considerations and other unobserved constraints that we intentionally exclude from our utility definition. Consequently, we do not aim to perfectly replicate historical acceptance rates (which we cannot replicate anyways due to limited data), but rather to isolate the trade-offs between this theoretical utility and fairness.

To quantify this utility, we construct a score u_i for each applicant based on available academic features: GPA (on a 4.0 scale) and GMAT (max 800 points). A simple construction is a normalized,

³In our project proposal, we also included the Law School Dataset (Wightman, 1998). However, upon further analysis, we discovered that it did not contain all the data we desired, and we abandoned the dataset.

equally weighted sum: $u_i = 0.5 \cdot (\text{GPA}_i/4.0) + 0.5 \cdot (\text{GMAT}_i/800)$. We conduct a brief robustness check of the sensitivity of our results to this particular choice (by varying the weights and including work experience) and find that our results do not depend much on this choice. We also note that the agreement between the baseline results given by our choice of the utility and actual admission results never surpasses 39%. Further details on this are in Appendix B.

Defining the proxy for ability: For Equality of Opportunity fairness (Section 3), we condition on the true ability of students a_i . However, most empirical datasets do not (and cannot) include such information. To address this, we create a proxy variable ($a_i = 1$ if $u_i \geq \text{median}(u)$). We acknowledge that this choice of a_i might lead to interpretation issues as both u_i and a_i rely on the same variables, however — given the limited amount of information available in our datasets — we couldn't find a practical way around this.

5.3 Solving & Metrics

To solve our optimization problem, we use the COIN-OR Branch and Cut solver (Forrest et al., 2005) with the python package PuLP (Mitchell et al., 2011) due to its versatility. We also use Gurobi (Gurobi Optimization, LLC, 2024) for larger problems, under its academic license. Even though we could instead solve cheaper LP relaxations as derived in Section 4, we stick with the IP solvers as the solving is generally quick.

We compute and visualize a version of the efficiency-fairness Pareto frontier via the "price of fairness", the percentage loss in institutional utility to achieve a given level of fairness. We also calculate the maximum intersectional disparity, $\max_h(s_h) - \min_h(s_h)$, where s_h is the selection rate of intersectional group h .

6 Empirical Results

We first present a fact about the dataset: even when only optimizing for institutional utility, it is (remarkably) fair. We present fairness rates of the unconstrained baseline (least fair) and the max-min Rawlsian solution (most fair) in Figure 1 and discuss this finding in Section 7.

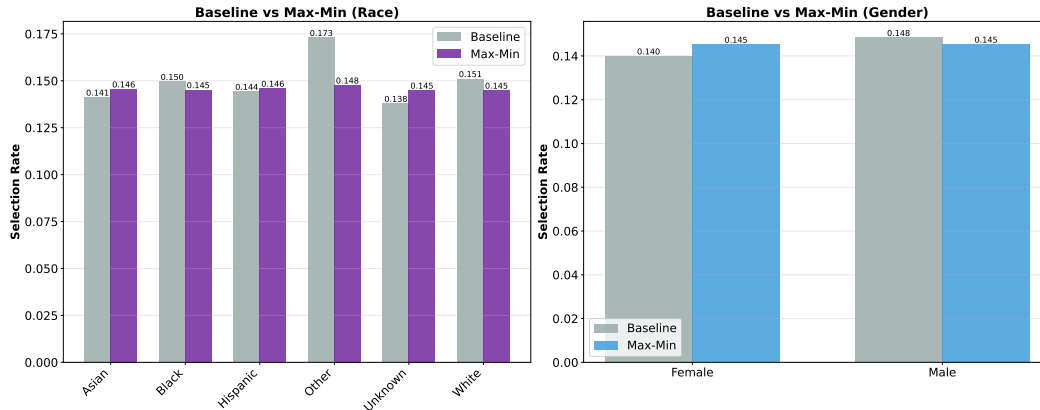


Figure 1: Baseline and Max-Min selection rates for each race (Left) and gender (Right). Note that the high selection rate for race "Other" might be an artifact of the dataset, as this group has a very small sample size (237 out of 6,194).

Selection Rates and Metric Incompatibility. We first examine how the composition of the admitted cohort shifts under strict fairness constraints. Figure 2 displays the selection rates for the Baseline model compared to Demographic Parity (DP), Equality of Opportunity (EO), Intersectional DP (IDP), and Max-Min Fairness.

The Baseline model (leftmost bars) exhibits noticeable disparity, particularly in the race categories where selection rates range significantly. Implementing Max-Min Fairness and IDP results in the

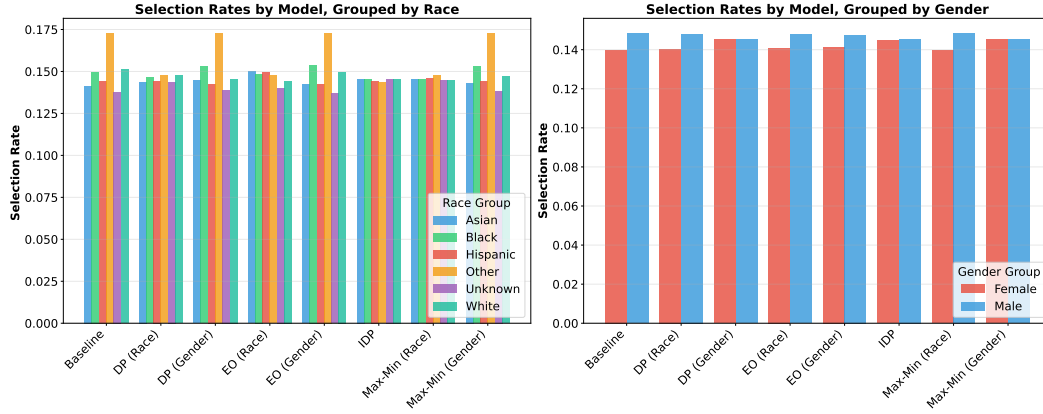


Figure 2: Selection rate by different fairness models and sensitive attributes. All selection rates are calculated under the smallest feasible ϵ .

most strictly egalitarian outcomes, effectively flattening the selection rates across groups to the global average.

A critical observation from Figure 2 is the phenomenon of *metric incompatibility*. When we enforce fairness on a single attribute, such as DP (Race), we achieve near-perfect equality across racial groups (Left plot). However, looking at the same model in the Gender grouping (Right plot), disparities between Male and Female remain. Conversely, DP (Gender) equalizes selection rates between men and women but leaves significant variance among racial groups. This confirms that fairness constraints usually operate orthogonally; satisfying one definition does not propagate fairness to other protected attributes. This is in particular true for intersectional fairness.

Intersectional Disparity Analysis. A key question is whether enforcing fairness on broad categories (Race and Gender) implicitly resolves intersectional disparities. Figure 3 compares the maximum intersectional disparity (the largest difference in selection rates between any two subgroups $h, h' \in H$) across models at their smallest feasible ϵ .

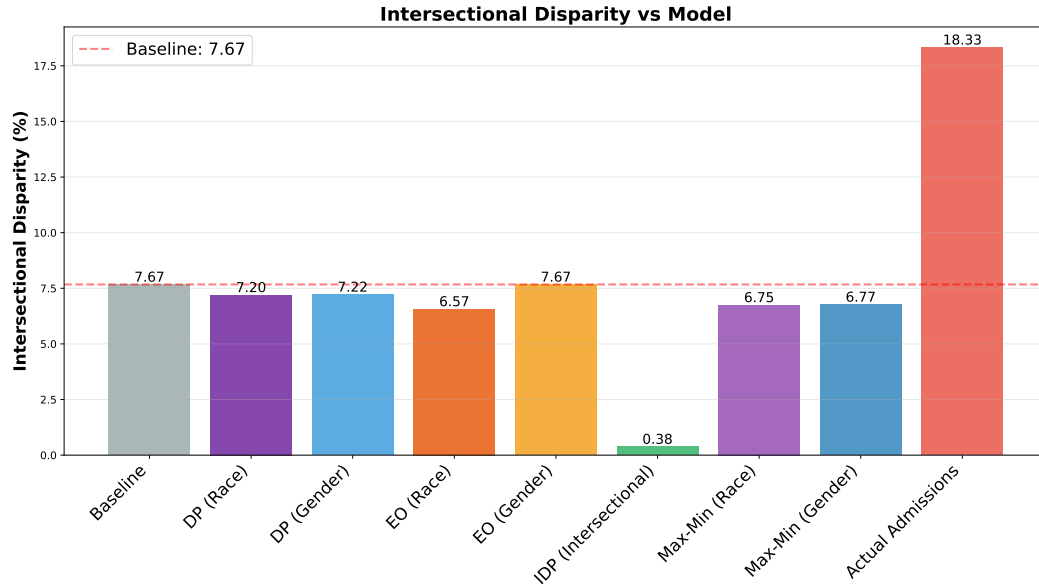


Figure 3: Maximum intersectional disparity across models. Only the IDP model successfully mitigates disparities between intersectional subgroups (e.g., Race $\times$ Gender combinations).

The Baseline model shows an intersectional disparity of 7.67%. Interestingly, the actual historical admissions data (far right) exhibits a massive disparity of 18.33%, suggesting our utility-based baseline is already "intersectionally fairer" than historical human decision-making (this does not necessarily hold for other fairness notions as well, such as gender parity).

Crucially, the single-attribute models (DP Race, DP Gender, EO) barely reduce intersectional disparity, maintaining levels around 6.5% – 7.2%. This indicates that fairness does not "trickle down"; ensuring parity for "Women" as a whole does not protect specific subgroups like "Hispanic Women." Only the IDP model successfully collapses this metric, achieving a negligible disparity of 0.38%.

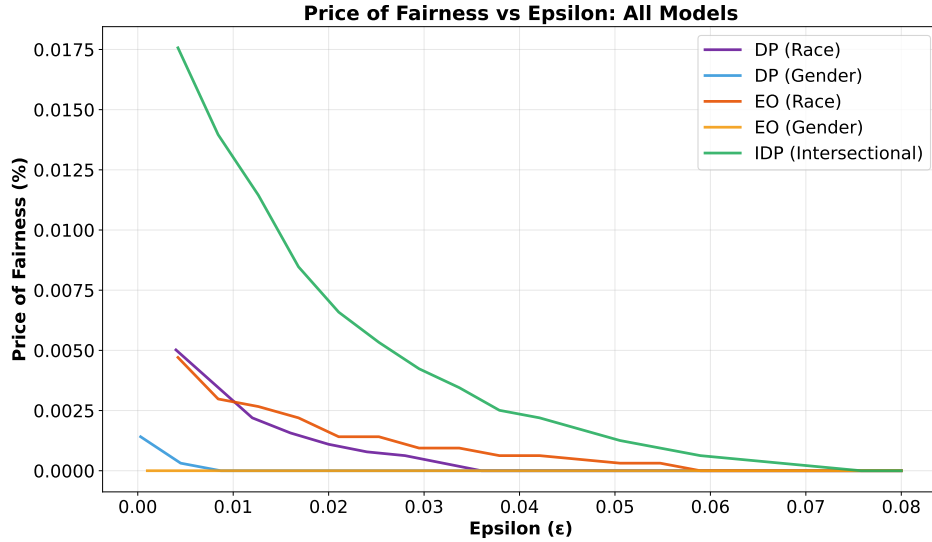


Figure 4: Price of fairness for DP, EO, IDP across race and gender.

Price of Fairness. In Figure 4, we compare the Price of Fairness (PoF) for a range of meaningful ϵ values, sweeping over both sensitive attributes and fairness notions (except Max-Min and α -Fairness, as those do not depend on ϵ). The curves plateau at $\epsilon \approx 0.0767$ (7.67%), which represents the natural disparity of the unconstrained baseline. As ϵ decreases (stricter fairness), the PoF increases.

Notably, the IDP (Intersectional) model is significantly more "expensive" than single-attribute models. This is expected, as IDP imposes constraints on a much larger number of smaller subgroups (e.g., Black Women, Asian Men), drastically reducing the solver's feasible space. We also note that the curves initiate after $\epsilon = 0$. As discussed in our Section 3, strict fractional constraints on small subgroups render the problem infeasible for very small ϵ due to granularity limits (i.e., one student represents a discrete jump in selection rate).

Axiomatic Fairness (α -Fairness). Lastly, we analyze the α -Fairness formulation, which modifies the objective function rather than imposing hard constraints. Here, we define the objective as maximizing the α -fair utility of groups (so that $\alpha = 0$ corresponds to our linear utility baseline).

Figure 5 (Left) demonstrates the "leveling" effect of increasing α . At $\alpha = 0$ (Baseline), there is a distinct gap between Male and Female total utility. As α increases, the solver prioritizes the lower-utility group (Female), effectively equalizing the total utility between genders. This convergence mirrors the behavior of Max-Min fairness. The trade-off is visible in the right plot: as α increases, the total utility (the sum of all utility scores) decreases.

7 Discussion

In this project, we investigated algorithmic fairness of university admissions. Our investigation reveals that while fairness constraints are effective at enforcing specific egalitarian goals, they are subject to significant trade-offs regarding feasibility, data quality, and the definition of institutional utility.

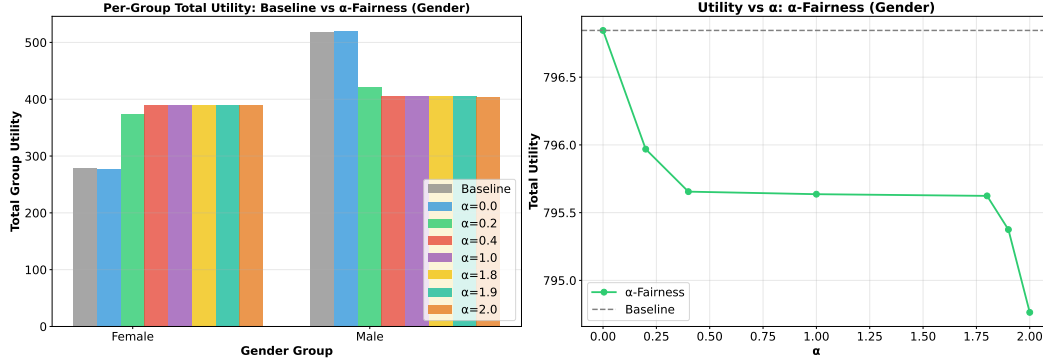


Figure 5: Impact of α -Fairness on group utility (Left) and total utility (Right). As α increases, the utility distribution becomes more equitable, approaching the Max-Min solution, but overall utility decreases. We approximate the nonlinear objective functions via piece-wise linear functions (PWL()) in Gurobi).

The Baseline Fairness Paradox. A somewhat counter-intuitive finding from our empirical analysis is the relative fairness of the unconstrained utility-maximising baseline. We believe two factors likely contribute to this. First, the context of MBA admissions differs significantly from undergraduate admissions; the applicant pool is older, strictly self-selecting, and often homogenized by work experience requirements, potentially narrowing the variance in academic metrics across demographic groups. Second, the actual admissions disparity likely reflects unobserved variables (e.g., essays, interviews, donor status) or implicit biases that are not captured in our constructed utility u_i or even in our data at all. Consequently, while the utility-maximizing baseline is not perfectly fair, it might not be the primary driver of the massive disparities often observed in higher education.

Nonetheless, it should be noted that actual admissions are still far from "fair": as shown in Figure 3, the baseline intersectional disparity ($\approx 7.67\%$) is considerably lower than that of the actual admissions decisions ($\approx 18.33\%$). This suggests that a pure meritocracy based strictly on the observed metrics (GPA and GMAT) is arguably fairer—in terms of intersectional selection rates—than the historical human decision-making process.

Feasibility and Computational Tractability. We identified the risk of finding "trivial or infeasible solutions" in our initial proposal, and this proved to be a material constraint. As predicted by our analysis of the model's integer constraints (Section 4), imposing strict fairness budgets ($\epsilon < 0.01$) often rendered the problem infeasible, particularly for Intersectional Demographic Parity.

Regarding computational intractability, we did not observe significant runtime bottlenecks, even without the reformulation of fairness constraints into cardinality bounds.

Interpretation of Utility Loss. The "Price of Fairness" (PoF) observed in Figure 4 remained surprisingly low, rarely exceeding a 0.01% reduction in total utility even under the strictest intersectional constraints. However, this metric requires a careful "sanity check": first, a 1% drop in total utility does not imply that only 1% of the cohort changed. Because the distribution of applicant utility scores is dense, the optimization model can swap a "rejected" candidate for an "accepted" candidate with only a marginally lower score (e.g., swapping a candidate with $u_i = 0.98$ for one with $u_i = 0.97$). This swap has a negligible impact on the objective function but a binary impact on the demographic composition.

Second, we interpret the magnitude of the 'Price of Fairness' with caution. Our measure of institutional utility is a dimensionless construct specific to our selection and weighting choices. Consequently, a numerically small percentage decrease does not necessarily imply that the cost of fairness is negligible in practice and does not necessarily map linearly to real-world outcomes – but then again, some might say the same is true for GPAs.

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A Derivation of Cardinality Bounds

In this section, we derive the integer cardinality bounds L_g and U_g used in our reformulated model. We show that these bounds guarantees that the Demographic Parity constraint is satisfied while preserving the total unimodularity (TU) of the constraint matrix.

Recall the original fractional constraint for Demographic Parity from Section 3, which requires that the difference in selection rates w_g between any two groups does not exceed ϵ :

$$|w_g - w_{g'}| \leq \epsilon \quad \forall g, g' \in G \quad (4)$$

where $w_g = \frac{1}{|N_g|} \sum_{i \in N_g} x_i$.

Let $r = k/n$ be the fixed global acceptance rate. To ensure pairwise disparity is bounded by ϵ , we impose a stricter condition: every group's selection rate must lie within a radius of $\epsilon/2$ from the global average r . By the triangle inequality, if $|w_g - r| \leq \epsilon/2$ and $|w_{g'} - r| \leq \epsilon/2$, then:

$$|w_g - w_{g'}| = |(w_g - r) - (w_{g'} - r)| \leq |w_g - r| + |w_{g'} - r| \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \quad (5)$$

Thus, creating bounds around the global average guarantees the original fairness requirement. We express this centered constraint as:

$$\frac{k}{n} - \frac{\epsilon}{2} \leq \frac{1}{|N_g|} \sum_{i \in N_g} x_i \leq \frac{k}{n} + \frac{\epsilon}{2} \quad (6)$$

To convert this into a linear constraint with integer coefficients (preserving the network matrix structure), we multiply the entire inequality by the group size $|N_g|$:

$$|N_g| \left(\frac{k}{n} - \frac{\epsilon}{2} \right) \leq \sum_{i \in N_g} x_i \leq |N_g| \left(\frac{k}{n} + \frac{\epsilon}{2} \right) \quad (7)$$

Since the number of admitted students $\sum x_i$ must be an integer, we tighten the bounds by taking the ceiling of the lower bound and the floor of the upper bound. This yields the final cardinality constraints L_g and U_g :

$$L_g = \left\lceil |N_g| \left(\frac{k}{n} - \frac{\epsilon}{2} \right) \right\rceil \quad (8)$$

$$U_g = \left\lfloor |N_g| \left(\frac{k}{n} + \frac{\epsilon}{2} \right) \right\rfloor \quad (9)$$

Enforcing $L_g \leq \sum_{i \in N_g} x_i \leq U_g$ ensures that the pairwise disparity never exceeds ϵ (subject to integer rounding), and leaves the LP relaxation integral.

B Robustness of the Utility Definition

To assess the sensitivity of our results to the specific choice of utility function, we conduct a comprehensive robustness analysis examining how different utility constructions affect both the baseline selection and the agreement with historical admission decisions.

B.1 Baseline Utility Sensitivity

Our primary utility specification uses an equally weighted combination of normalized GPA and GMAT scores: $u_i = 0.5 \cdot (\text{GPA}_i/4.0) + 0.5 \cdot (\text{GMAT}_i/800)$. To test sensitivity to this weighting scheme, we systematically vary the relative weights between GPA and GMAT from (0.0, 1.0) to (1.0, 0.0) in increments of 0.1, maintaining the constraint that weights sum to unity. For each weight combination, we solve the unconstrained utility maximization problem and compute the overlap percentage—defined as the fraction of selected candidates who were also admitted in the historical data.

The results demonstrate that the overlap with actual admissions is relatively stable across weight combinations, ranging from approximately 30.9% (when using only GPA) to 39.0% (with equal weights). The equal-weight specification (0.5, 0.5) achieves the maximum observed overlap of 39.0%, corresponding to 351 out of 900 selected candidates matching historical admissions. This finding suggests that neither GPA nor GMAT alone captures the full selection criteria used in practice, and that the balanced combination provides the best available proxy given our observable features.

B.2 Extended Utility Definitions

Even though we cannot and do not seek to predict actual admissions, we further examine whether incorporating additional observable features can improve agreement with historical decisions. Specifically, we construct an extended utility function that includes normalized work experience (using a saturating transformation $1 - e^{-w/3}$ where w is years of experience), work industry bonuses (assigning higher weights to industries such as Consulting, Investment Banking, and Private Equity) and major-specific bonuses (STEM fields receive a 10% bonus, Business fields receive a 5% bonus, and Humanities receive no bonus), with weights that we optimize. These bonuses are entirely heuristic and not based on empirical evidence; they serve only as a rough approximation of the kinds of preferences that MBA admissions committees might plausibly apply.

Despite these extensions, the overlap with actual admissions does not meaningfully improve. Sensitivity analysis across 13 different weight combinations for the extended utility (varying the relative importance of GPA, GMAT, work experience, work industry and major) reveals that overlap percentages remain tightly clustered between 35.4% and 39.1%, with the highest value (39.1%) achieved when work experience and major bonuses are nearly eliminated (weights: 0.45 GPA, 0.45 GMAT, 0.1 work experience, 0 major, 0 work industry), effectively returning to a GPA–GMAT only specification.

B.3 Implications

These findings have two main implications. First, our fairness–utility trade-off results are robust: changing the utility definition has little effect on baseline selection patterns or the shape of the Pareto frontier, suggesting that our conclusions do not depend on any particular choice of weights or features.

Second, the relatively low agreement with historical admissions (never above 39%) indicates that real decisions relied on factors not present in our dataset—such as personal statements, recommendations, extracurriculars, or diversity considerations. This reflects a limitation by design: focusing on observable academic metrics allows us to isolate the theoretical trade-off between merit and fairness.

C Data Limitations and Ability Proxies

Finally, our analysis was constrained by the quality of available data. We initially intended to utilize the Law School Admissions dataset to construct a ground-truth proxy for ability (a_i) based on bar exam passage rates. However, data quality issues necessitated a reliance on the synthetic Wharton dataset.

This limitation most severely impacts the *Equality of Opportunity* (EO) formulation. In our model, the qualification proxy a_i was derived from the same features (GPA/GMAT) used to calculate utility u_i . As a result, a_i and u_i were highly correlated, making the EO constraint behave similarly to the utility-maximising baseline. A dataset containing distinct outcome measures (e.g., post-graduation salary, job placement) would allow for a more robust separation between "admissions utility" and "true qualification".